

Strict-Causal R-Peak Prediction: a PI phase tracker with a scalar-Kalman gain

Implementation & analysis note (CFA ear-EEG)

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Scope. Strict-causal cancellation subtracts the template at a *predicted* cardiac phase, because the current beat's R-peak has not arrived. This note specifies the predictor: a PI phase-locked tracker whose integral (rate) gain is the steady-state Kalman gain for a scalar random-walk model of the RR interval. The same model yields the one-step prediction variance, which sets the strict-causal R-spike suppression ceiling and an adaptive fall-back-to-buffered gate. Assumes R-peak *detection* is already reliable (locking succeeds); this is purely the *prediction* problem.

1 What must be predicted

Between beats we advance a phase estimate and look up the template at that phase. At sample n , after the last detected R-peak $\hat{t}_k$, the strict-causal phase is

$$\phi_{\text{pred}}(n) = \frac{n - \hat{t}_k}{\hat{T}_k f_s}, \quad \hat{s}(n) = y(n) - \hat{d}_r(\phi_{\text{pred}}(n)), \quad (1)$$

using only the current RR estimate $\hat{T}_k$ (samples), no future fiducial $\Rightarrow$ zero added latency. Two quantities drive the error: the *onset* (where the next R falls, sets QRS-spike alignment) and the *rate* $1/\hat{T}_k$ (sets phase advance, dominates the T-wave tail under HRV). Both reduce to one estimation problem: predict the next RR interval.

2 Scalar RR state-space model

Model the beat interval T_k as a slowly-drifting scalar state with detection-jitter measurement (Assumption 3):

$$T_k = T_{k-1} + w_k, \quad w_k \sim (0, q) \quad (\text{process: RR drift / RSA}), \quad (2)$$

$$z_k = T_k + v_k, \quad v_k \sim (0, r) \quad (\text{measurement: } z_k = \hat{t}_k - \hat{t}_{k-1}). \quad (3)$$

q is the beat-to-beat RR increment variance (how fast the rate wanders), r the R-peak detection timing-jitter variance. Both are *measurable from data*: q from the variance of successive RR differences, r from fiducial jitter against a clean reference.

3 Kalman filter and the steady-state gain

The scalar Kalman recursion for (2)–(3):

$$\text{predict: } \hat{T}_k^- = \hat{T}_{k-1}, \quad P_k^- = P_{k-1} + q, \quad (4)$$

$$\text{gain: } K_k = \frac{P_k^-}{P_k^- + r}, \quad (5)$$

$$\text{update: } \hat{T}_k = \hat{T}_k^- + K_k (z_k - \hat{T}_k^-), \quad P_k = (1 - K_k) P_k^-. \quad (6)$$

At steady state $P_k^- \rightarrow P^-$ solves the scalar Riccati $P^- = (1 - K)P^- + q$ with $K = P^-/(P^- + r)$, i.e. $P^{-2} - qP^- - qr = 0$, giving the closed form

$$\boxed{P^- = \frac{q + \sqrt{q^2 + 4qr}}{2}, \quad K_{ss} = \frac{P^-}{P^- + r}} \quad (7)$$

K_{ss} depends only on the ratio $\gamma = q/r$: with $\beta \equiv P^-/r = \frac{1}{2}(\gamma + \sqrt{\gamma^2 + 4\gamma})$, $K_{ss} = \beta/(\beta + 1)$. Large γ (rate wanders fast relative to jitter) $\Rightarrow K_{ss} \rightarrow 1$ (trust each measurement); small $\gamma \Rightarrow$ heavy smoothing.

4 PI phase tracker, and why its gain is K_{ss}

The deployed tracker is a PI phase-locked loop. Between beats the phase accumulates at the estimated rate; at each fiducial a proportional term corrects onset phase and an integral term updates the rate:

$$\text{advance: } p(n+1) = [p(n) + f_r(n)] \bmod 1, \quad f_r(n) = \frac{1}{\hat{T}_k f_s}, \quad (8)$$

$$\text{onset (P): } p \leftarrow p - K_p e_p, \quad (9)$$

$$\text{rate (I): } \hat{T} \leftarrow \hat{T} + K_i (z_k - \hat{T}). \quad (10)$$

The integral update (10) is *identical* to the Kalman state update with

$$\boxed{K_i = K_{ss}} \quad (11)$$

So the PI form is not an ad-hoc alternative to a Kalman filter — it *is* the steady-state scalar Kalman filter, with the integral gain *derived* from the measured RR statistics (7) rather than hand-tuned. The proportional gain K_p governs onset (fiducial-phase) correction, separate from rate: $K_p \rightarrow 1$ snaps phase to the measured fiducial, $K_p < 1$ smooths onset jitter. Set the loop bandwidth above the RSA band (~ 0.2 – 0.4 Hz) and below per-beat detection noise.

Design recipe. (1) estimate q, r from data; (2) $\gamma = q/r \Rightarrow K_i = K_{ss}$ via (7); (3) pick K_p for onset smoothing (start $K_p \approx 1$, reduce if fiducial jitter dominates). No loop to hand-tune; the one number γ sets the rate gain.

5 One-step prediction variance $\rightarrow$ R-spike suppression cap

Strict-causal mode predicts the next R-onset as $\hat{t}_{k+1}^{\text{pred}} = \hat{t}_k + \hat{T}_k$. The prediction error is $\delta\tau = \hat{T}_k - T_{k+1} = (\hat{T}_k - T_k) - w_{k+1}$, whose variance is exactly the steady-state *prior*:

$$\sigma_\tau^2 = \text{Var}[\delta\tau] = P^+ + q = P^-, \quad \sigma_\tau = \sqrt{P^-}. \quad (12)$$

The Kalman model therefore hands you the prediction jitter σ_τ directly. Because the R-spike is the sharpest template feature (width w_R), a lookup misaligned by σ_τ attenuates its peak by approximately a Gaussian-blur factor,

$$\frac{\hat{d}_r^{\text{causal}}(0)}{\hat{d}_r^{\text{buffered}}(0)} \approx \exp\left(-\frac{1}{2}(\sigma_\tau/w_R)^2\right), \quad (13)$$

while the broad QRS–T tail (slow on the scale of σ_τ) is largely preserved. This closes the loop: $\{q, r\} \Rightarrow \sigma_\tau \Rightarrow$ predicted strict-causal R-spike ceiling, computable *before* deployment. (Placeholder: with $q = [\dots]$, $r = [\dots]$ one gets $\sigma_\tau = [\dots]$ ms and a predicted R-spike retention of $[\dots]\%$, to be checked against §7.)

6 Adaptive gating (arrhythmia-aware, in estimator form)

The running P_k^- inflates on irregular beats (large innovation $z_k - \hat{T}_k^-$ raises the model’s uncertainty), so per-beat $\sigma_{\tau,k} = \sqrt{P_k^-}$ is an online confidence. Gate on it:

$$\text{if } \sigma_{\tau,k} > \sigma_{\max} : \text{ fall back to buffered (one-beat delay) or freeze update.} \quad (14)$$

This is the arrhythmia supervisor expressed through the estimator: strict-causal when prediction is confident, buffered when it is not. Choose $\sigma_{\max}$ as a fraction of w_R (e.g. the σ_{τ} at which (13) drops below an acceptable R-spike retention).

7 Optional: RSA structure (only if the residual warrants it)

The random-walk (2) is memoryless. If the RR-prediction residual is *white*, it is provably optimal and nothing fancier is justified — check this first. If the residual is structured at the respiratory frequency, replace (2) with an AR(1)-toward-mean or a two-state (RR + respiratory-phase) model; the Kalman machinery is unchanged, only the state transition grows. Do not add the respiratory state on intuition — add it only if the whiteness test on the residual fails.

8 Validation protocol (the reviewer’s jitter sweep doubles as predictor test)

Inject controlled R-peak prediction jitter $\sigma_{\tau} \in \{0, 2, 5, 10, 20, 40\}$ ms and plot R-spike (and cardiac-band) suppression vs σ_{τ} for three anchors: (i) oracle (true next R, upper bound), (ii) PI/Kalman-predicted, (iii) buffered (one-beat delay). This simultaneously validates the predictor against the oracle ceiling, confirms the cap formula (13) on data, and locates the crossover where buffered beats strict-causal — i.e. where the gate §6 should switch.

Algorithm summary (per sample n ; beat index k)

once, offline: estimate q, r from data; $K_i = K_{ss}(q, r)$ via Eq. (kss); pick K_p , $\sigma_{\max}$
state: p (phase), RR_{hat} (samples), P (RR prior var), last_fid t_hat_k

on each sample n:

p_pred = (n - t_hat_k) / (RR_hat)	# Eq. (pred), fractional phase
idx = round(p_pred * RR_hat)	# template offset from R
s_hat[n] = y[n] - d_hat[idx]	# strict-causal cancellation
p += 1.0 / RR_hat	# advance phase

on detected fiducial at sample n (beat k):

z = n - t_hat_k	# measured interval
innov = z - RR_hat	
P = P + q	# predict
K = P / (P + r)	# (== K_i at steady state)
RR_hat = RR_hat + K * innov	# rate update Eq. (I)
P = (1 - K) * P	
p = p - K_p * wrap(p)	# onset correction Eq. (P)
sigma_tau = sqrt(P + q)	# one-step pred jitter Eq. (sigtau)
if sigma_tau > sigma_max: mode = BUFFERED	# gate Sec. 6
else: mode = STRICT_CAUSAL	
t_hat_k = n	

Constant work per sample ($O(1)$), one template of length n_t in memory ($O(n_t)$). The only tuned quantities are $\{q, r\}$ (measured) and $\{K_p, \sigma_{\max}\}$ (onset smoothing / gate); the rate gain follows from (7).